

Perdiction Results

2024-10-02

Data Summary:

- **Number of observations for bill passage predictions:** 40,000. Initially, 4 requests returned null responses, but upon retrying with the same prompts and removing `max_token=null`, they returned non-null responses.
- **Number of observations for bill summary completion:** 39,999. A total of 57 responses exceeded the token limit of 500, and 1 response was dropped due to content filtering.

Results

Bill Passage

Table 1: True Bill Passage Rates

Chamber	True Passage Rate
Pass House	0.074
Pass Senate	0.055

Table 2: Performance Evaluation of LLM Predictions of Bill Passage

Predicted Outcome	Model	Prompt	Accuracy	TPR	FPR
Pass House	GPT-3.5	w/o date restriction	0.223	0.974	0.837
Pass House	GPT-3.5	w/ date restriction	0.230	0.972	0.829
Pass House	GPT-4o	w/o date restriction	0.914	0.127	0.023
Pass House	GPT-4o	w/ date restriction	0.709	0.566	0.280
Pass Senate	GPT-3.5	w/o date restriction	0.316	0.942	0.721
Pass Senate	GPT-3.5	w/ date restriction	0.312	0.940	0.724
Pass Senate	GPT-4o	w/o date restriction	0.934	0.176	0.022
Pass Senate	GPT-4o	w/ date restriction	0.761	0.640	0.232

Bill Summary Completion

Table 3: Performance Evaluation of LLM Completion of Bill Summaries (Non-Embedded)

Model	Prompt	Average
GPT-3.5	w/o date restriction	0.015
GPT-3.5	w/ date restriction	0.016
GPT-4o	w/o date restriction	0.039
GPT-4o	w/ date restriction	0.035

Table 4: Performance Evaluation of LLM Completion of Bill Summaries (Embedded)

Metric	Model	Prompt	Average	Baseline	Relative to Baseline ($\uparrow$ Similar)
Cosine Similarity	GPT-3.5	w/o date restriction	0.488	0.245	0.996
Cosine Similarity	GPT-3.5	w/ date restriction	0.497	0.243	1.044
Cosine Similarity	GPT-4o	w/o date restriction	0.533	0.242	1.201
Cosine Similarity	GPT-4o	w/ date restriction	0.538	0.243	1.211
Euclidean Distance	GPT-3.5	w/o date restriction	0.987	1.227	0.196
Euclidean Distance	GPT-3.5	w/ date restriction	0.977	1.228	0.205
Euclidean Distance	GPT-4o	w/o date restriction	0.925	1.229	0.248
Euclidean Distance	GPT-4o	w/ date restriction	0.922	1.228	0.249